
Pruning vs Quantization: Which is Better?

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Abstract

Neural network pruning and quantization techniques are almost as old as neural networks themselves. However, to date only ad-hoc comparisons between the two have been published. In this paper, we set out to answer the question on which is better: neural network quantization or pruning? By answering this question, we hope to inform design decisions made on neural network hardware going forward. We provide an extensive comparison between the two techniques for compressing deep neural networks. First, we give an analytical comparison of expected quantization and pruning error for general data distributions. Then, we provide lower bounds for the per-layer pruning and quantization error in trained networks, and compare these to empirical error after optimization. Finally, we provide an extensive experimental comparison for training 9 large-scale models on 4 tasks. Our results show that in most cases quantization outperforms pruning. Only in some scenarios with very high compression ratio, pruning might be beneficial from an accuracy standpoint.¹

1 Introduction

Recent advances in deep learning led to exceeding human-level performance in many tasks, including computer vision, machine translation, voice recognition, and language understanding. Real-world applications of DNNs rely heavily on their efficiency. Both mobile and cloud platforms greatly benefit from reduced latency and energy efficiency achieved by some form of model compression. In this work, we consider two mainstream techniques used in practice; pruning and quantization.

Pruning methods remove individual weights [70, 25], or sometimes groups of weights [28, 47]. This procedure can reduce the memory footprint. Furthermore, not having to perform the computations with weights that are zeroed out can make network inference more efficient. On the other hand, quantization reduces the bit-width used for both the weights and the computation used in networks, leading to both predictable memory savings and reductions in the necessary compute. In both scenarios, the hardware used for making use of these optimization schemes needs to take them into account.

Depending on the availability of training data and computing budget, most methods for pruning and quantization fall into one of two families. The first family includes fine-tuning approaches, namely quantization-aware training (QAT) and fine-tuning with pruning in the loop. The second family includes post-training approaches such as post-training quantization (PTQ). Previously, pruning techniques primarily relied on fine-tuning; however, some post-training pruning methods appeared recently as fine-tuning is not desirable for large language models [18].

Despite the importance of model efficiency and the plethora of approaches for pruning and quantization, the two fields are mostly disjoint. The literature presents little insight into which of the two

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¹Code is available at <https://github.com/Qualcomm-AI-research/pruning-vs-quantization>